

Chapter – 9: Methods of Separation in Everyday Life

Just like winnowing makes the husk blow away while retaining the grains, similarly, sages let the virtues stay while discarding the vices. – Kabir

- Malli and sister Valli – excited – summer vacation
- Their parents – planned trip – across India – meeting relatives
- 1st halt – *Nani's* (maternal grandmother) house – Haryana – surrounded by – large fields
- Malli, Valli – fascinated – variety of grains – courtyard
- *Mami* (maternal aunt), *mama* (maternal uncle), other community members – separating small stone, husk – hands
- Malli, Valli – curious – why doing it?
- *Nani* – notice curiosity – tells them – stones removed – make grains – fit for cooking
- Go around house – see fields, cattle
- Keep them busy – *nani* – offers challenge – handpick stones – eyes closed
- Method – picking by hand – unwanted substance from **mixture** (2 or more substances mixed together) – **handpicking** – ex. – picking small stones, husk – wheat, rice
- Basis – color, size, shape – diff.
- Particles to be removed – small qty. – can be picked by hand – very convenient method
- Lunch – Malli, Valli – served – hot vegetable pulao
- *Nani* – obs. – Malli – separating – black peppers – Valli teases – nice handpicking
- *Nani* – teaches – benefits of black pepper – encourages to eat
- Later - *mama* – take them – fields – they **obs.** – bundles – harvested wheat stalks – lying – some spread in Sun – drying
- Both – pick 1 stalk each – notice – numerous grains – attached
- Grp. – farmers – beating stalk – wooden log
- Valli – curious – asks *mama* – why doing it?
- He explains – beating stalks – separate grains
- This process – separating grains – beating stalks – **threshing**
- Farmers – work hard – enjoy work – sing songs in between
- Separated grains – mixed with – piles of husk
- Valli whispers – will farmers – handpick them – how much time?
- Activity –
 - Take – handful peanuts – rub them – b/w palms
 - Try – separating – removed skin and peanuts – little hard
 - Try – blowing it – obs.
- Blowing air – separate heavier and lighter components
- Traditionally – *soop* (bamboo tray) – separate heavier and lighter components
- Next day – *nana* – take them – fields – show this process
- Malli, Valli – obs. – farmer – raised platform – moving bamboo tray – direction of air
- Grains – heavier – fall directly below – husk – lighter – falls far away
- This method – separating heavier, lighter components – using wind – **winnowing**
- Next day – Malli, Valli – board train – Ahmedabad – visit – father's friend – Ghanshyam bhai
- Before leaving – Valli asks *mami* – make *meethi puri*
- Valli – can I help – knead – wheat flour
- *Mami* – prepare dish – flour – 1st – remove bran

- Valli – how to do it?
- *Mami* – use sieve
- **Sieving** – fine flour particles – pass through holes – bigger particles – bran, stones – remain on sieve
- Obs. – sieve – all holes – same size
- Sieving – used – solid-solid mixture – components – diff. size
- Reaching Ahmedabad – visit Sabarmati Ashram – with Ghanshyam bhai – learn – *Namak Satyagrah* (Dandi march)
- Malli – *namak* (common salt) – obtained where?
- Ghanshyam bhai – oceans, seawater
- Seawater – mixture – salts, other substances – dissolved in water
- Obtain salt – seawater – pour – shallow pits – exposed – sunlight
- Few days later – water **evaporates** – leave behind – solid mixture
- Common salt – this mixture – further purification
- Activity –
 - Take – bowl – fill half – water
 - Add – 2-3 teaspoons – salt – stir – until salt dissolves
 - Take – small piece – thick black paper – spread few drops – salt solu.
 - Create any art – your choice
 - Allow – dry – obs.
- Touch paper – feel presence – salt
- Water – disappeared – soaked by paper
- Activity –
 - Take – salt solution – previous act. – china dish
 - Heat – water boils away – allow – cool down – obs.
- Touch china dish – feel presence – salt
- Now Malli, Valli – visit *dada, dadi* (paternal grandparents) – Puducherry – southern India
- Excited – meet old friend – Balan
- Reach Puducherry – busy talking – time passes – evening already – *dadi's* tea time
- *Dada* – prepares tea
- Children – helps him
- *Dada* – shares tips – make tea
- Balan – after preparing – how to remove – tea leaves
- *Dada* – with a strainer – BUT – without it can also be done
- Valli – how?
- *Dada* – leave – sauce pan – tea inside – undisturbed – gently pour tea – in cup
- Valli – yes – tea leaves – settled – bottom
- Process – settle down – heavier insoluble components – bottom of liquid – **sedimentation**
- Water – removed – tilting vessel – process – **decantation**
- *Dada* – decantation – not 100 % accurate – still get some tea leaves
- Balan – this method – not proper
- *Dada* – tea is ready
- Malli – picks strainer – hand over to *dada*
- *Dada* – pours tea – through strainer – tea leaves – collected in strainer
- This process – separating tea leaves – **filtration**
- Balan asks Malli – use tea strainer – filter muddy water
- *Dada* – try piece of cloth – obs. diff.

- Malli – why piece of cloth?
- *Dada* – cloth – small holes – b/w threads – these pores – used as filter – people – ancient times – used this
- Water – still muddy – impurities – separate using filter – smaller holes – ex. – filter paper
- Activity –
 - Try – fold – filter paper – make – cone
 - Place it – funnel – conical flask – pour muddy water
 - Water – through funnel – collected – flask
 - Mud particles – residue – filter paper – clear water – filtrate
- Malli, Valli – boat ride – nearby river – *dada* and his friend – Otukkam
- They cast – fishing net – water drains
- Valli – recalls – this method – similar – filtration
- Malli – very upset – plastic bags, broken bottles, food wrappers, along with fish – inside net
- Malli, Valli – take blessings *dada*, *dadi* – say goodbye – Balan – travel – Madhya Pradesh
- Train – Bhopal – sun is shining – getting hot
- On the way – *maasi*’s (maternal aunt) home – drink *chhach* (buttermilk) – *dhaba* (roadside eatery)
- Malli – asks shopkeeper – big painting – wall
- He explains – picture – lady – performing – **churning** curd – using – big *mathni* (churner) – separate butter
- This process – butter – lighter – floats atop – buttermilk – left behind
- Their stay – *maasi*’s place – very enjoyable – excited to share exp.
- Now – time for final destination – Shillong – capital of Meghalaya
- On arrival – *bua*’s (paternal aunt) – house – Shillong – notice – carpenter – wooden door
- While working – accidentally – drops iron nails – saw dust
- Starts handpicking – children – stops him – get magnet – *bua*
- Ask carpenter – move magnet – saw dust – iron nails – stick to it
- Substances – attracted to magnet – magnetic substances – iron – common example
- Separation – magnetic, non-magnetic substances – using magnet – **magnetic separation**
- Malli, Valli – delightful holidays, *Bharat Yatra* (India tour) – very fun and knowledgeable